Topic 9.4 Practical circuits

Exercise 9.4A Variable resistors

1 

2 1 – 50 Ω

3 **a** current will decrease
b brightness will decrease

Exercise 9.4B Uses of variable resistors

1 The resistance of a variable resistor can be changed. The resistance of a fixed resistor cannot be changed. (Descriptions can be given in terms of values in ohms.)

2 Circuit diagram should have a cell, a lamp and a variable resistor in series.

3 Circuit diagram should have a cell, two lamps and a variable resistor in series.

4 Circuit diagram should have a cell, a variable resistor close to the cell, and two lamps in parallel (variable resistor in unbranched part).

Exercise 9.4C Comparing circuits

1 C

2 B

3 D

4 B

5 A, C and D

6 B and D